Public health data analysis

What is your expectations?

What is your expectations?

Fears?

Backround

- Mine
- Yours?
- Why do we have this course?

- Warnings!
  - Time schedule content is not fixed
  - I do not know whether I have made a really difficult course....

A look into the future….

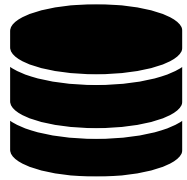
- Common feelings when learning R (and working with R):
  - Confusion
  - Frustration
  - Anger
  - The joy and relief of finally getting it right?
  - Sense of mastery?

How we normally work when doing analyses



Ask

Turn a broad concern into an answerable research question(s).



Prepare

Understand structure, quality and limitations.



Analyse

Choose methods and visualisations that fit the question.



Communicate

Explain results, uncertainty and relevance.

Organization of teaching

1. Start with a problem

A public-health question gives the analysis a purpose.

2. Teacher demonstrates

The teacher works through one defensible approach and explains the decisions.

3. Code along

You write and run the same code

4. Guided practice

You solve a similar problem. The teacher acts as a tutor.

5. Apply independently

In the home assignment you should find a problem and address it from start to finish.

Question → Data → Analysis → Interpretation

AI

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  - Some good
  - Some really bad

  →If you have no idea about what the code does, how can you know that it is correct?

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- Have I used AI to write code for this course? Yes, of course.

Tips for AI use

- Try to write the code yourself first
- Instruct the AI to be a tutor, not only to provide you with complete code
- Use AI to troubleshoot
  - Why does not this code work?
  - What does this error-message mean?
- Use it to find functions and solutions. Describe what you want to do
- Try to read, remember and write. Not copy-paste
- Have a look at the documentation of the function suggested

You should after the course have the skill

- Work independently and responsibly with health data in accordance with ethical and legal requirements

- Act on medical and health research
- Regional comitees for medical and health research ethics
- GDPR
- Sikt – Norwegian Agency for Shared Services in Education and Research
  - Data protection impact assessment (DPIA)
  - Data managment plan
  - Data storage

Regional Etisk Komité

- **Prior approval**
- **Assessment of whether the project must be submitted:** Submit the project if you are uncertain
- **Freely given, informed consent**
  - As a general rule, consent must be obtained.
  - For consent to be valid, it must be **voluntary, specific, informed and unambiguous**.
  - There must be no doubt that the individual has consented to participate in the research.
  - The General Data Protection Regulation does not require consent to be given in writing or orally. However, you must be able to document or demonstrate that consent has been obtained. Consent must be given through a clear affirmative action, such as providing a signature, clicking a button or ticking a box. **Written consent is recommended.**

Exception: Helseregisterlovern

- https://lovdata.no/dokument/NL/lov/2014-06-20-43

...the processing of data in health registries in which names, national identity numbers, or other personal identifiers may be processed without the consent of the registered individual (data subject)...

My translation

GDPR

Examples of the type of information that may allow the direct or indirect identification of an individual, and **therefore qualify as personal data**, are:

- name, surname, phone numbers of clients, stakeholders, employees, providers;
- identification numbers, such as an individual's client number, an individual's employee number,
- a booking reference;
- email addresses, location data;
- an individual's browsing history;
- an individual's purchase history and receipts;
- photos, videos and audio recordings containing images or sounds of individuals.
- With this personal data, an individual may be identified directly or indirectly:
- if your organisation is processing an individual's name or surname for example, this personal data allows the direct identification of this individual;
- if your organisation is processing an individual's client number or booking reference for example, this personal data may allow the indirect identification of that individual.
- any type of information processed in relation to the individual directly or indirectly identified (i.e. preferences, habits) will be considered as personal data as well.

GDPR: special categories of personal data

- an individual's health;
- an individual's sex life or sexual orientation;
- an individual's racial or ethnic origin;
- an individual's political opinions, religious or philosophical beliefs;
- an individual's biometric and genetic data;
- trade union membership.

- processing’ means any operation or set of operations which is performed on personal data or on sets of personal data, whether or not by automated means, such as collection, recording, organisation, structuring, storage, adaptation or alteration, retrieval, consultation, use, disclosure by transmission, dissemination or otherwise making available, alignment or combination, restriction, erasure or destruction;

→ What we do in this course

We can still analyze personal data, but..

- Data management plan
- DPIA: Data Protection Impact Assessment

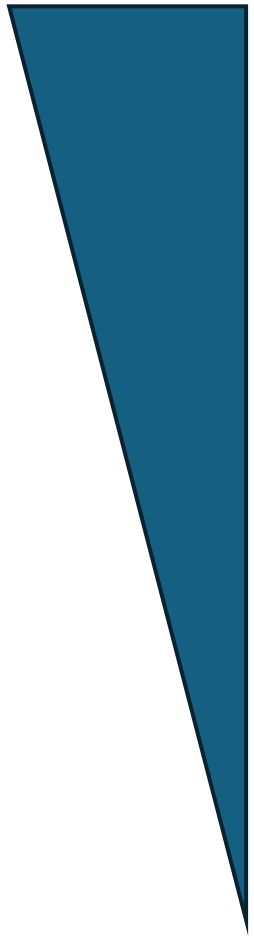
- To learn more
  - Data protection basics | Data protection guide for small business | European Data Protection Board

We need data to analyse!

- Collect?
- Research data?
  - Tromsø Study, NOWAC
  - Need application. Comes with a cost
- Publicly open research data.
  - NHANES, European social survey
  - Free of use with terms
- Public data
  - WHO
  - Statistics Norway. Other countries?
  - Other open sources?

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Regulatory demands
Data protection
Access
Cost

Software

- R
- Matlab
- Python

→To some degree similar

→And for what we are doing if you can describe what you want to do. Internet and LLM can help you find a way to do it.

Install R and Rstudio!

Backup today: Run R code online